

Creating Suspenseful Stories: Iterative Planning with Large Language Models

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Abstract

Automated story generation has been one of the long-standing challenges in NLP. Among all dimensions of stories, *suspense* is very common in human-written stories but relatively under-explored in AI-generated stories. While recent advances in large language models (LLMs) have greatly promoted language generation in general, state-of-the-art LLMs are still unreliable when it comes to suspenseful story generation. We propose a novel iterative-prompting-based planning method that is grounded in two theoretical foundations of story suspense from cognitive psychology and narratology. This theory-grounded method works in a fully zero-shot manner and does not rely on any supervised story corpora. To the best of our knowledge, this paper is the first attempt at suspenseful story generation with LLMs. Extensive human evaluations of the generated suspenseful stories demonstrate the effectiveness of our method.

1 Introduction

Among numerous NLP tasks, automated story generation is a representative one that requires creativity. By learning from human-written stories, an automated storyteller mimics humans and becomes competent in producing stories useful for various application scenarios, such as entertainment, education, and social bonding (Riedl and Young, 2010). The notable achievements in the field of deep learning have led to the subsequent emergence of data-driven methodologies for automated story generation (Martin et al., 2018; Clark et al., 2018; Fan et al., 2018, 2019; Yao et al., 2019). With the rapid development of large language models (Radford et al., 2019; Brown et al., 2020; OpenAI, 2022), generated stories have further increased greatly in length, complexity, and fluency. These enhancements are primarily realized through the application of methodologies that are built based

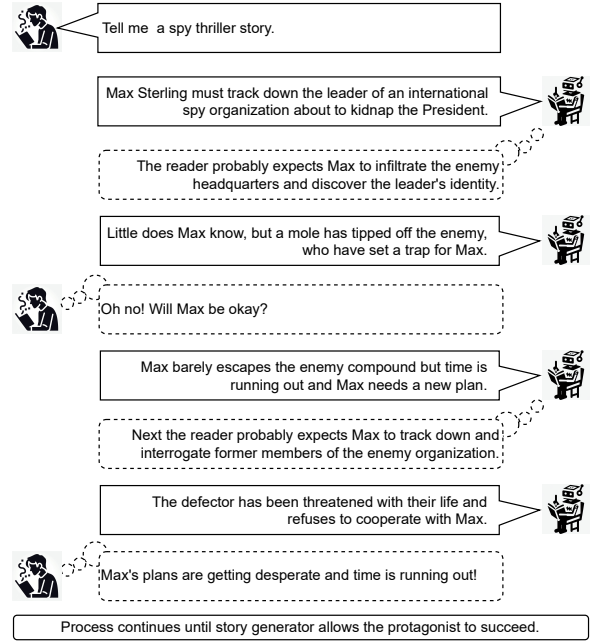


Figure 1: Our suspenseful story generation technique is predicated on a theory of suspense in which the quantity and quality (likelihood) of ways in which a protagonist can avoid an undesirable outcome are decreased. Our method iteratively produces possible actions for the protagonist to take and adversarially creates the conditions under which the protagonist will fail. The dialogue boxes indicate the story snippets generated by our method. The thought bubbles of the reader represent the mental process. The thought bubbles of the system depict the planning process in our method.

on either fine-tuning pre-trained models on supervised story datasets or prompting large language models (LLMs) of strong zero-shot capability.

Existing work on language model based computational storytelling mostly focuses on optimizing automated story generation systems from different angles, including both generation settings (e.g., long-form generation (Goldfarb-Tarrant et al., 2020; Yang et al., 2022b,a), controllable generation (Peng et al., 2018; Lin and Riedl, 2021; Peng et al., 2022)), and story characteristics (e.g., com-

nonsense reasoning (Guan et al., 2019, 2020; Ammanabrolu et al., 2021), temporal and causal relationships (Guan et al., 2020; Ammanabrolu et al., 2021; Han et al., 2022)).

In this paper, aiming to expand the capability of automated storytellers, we direct our focus towards an under-explored dimension of stories: *suspense*, which is one of the strong affective responses humans often feel when being told a story. Suspense is important as expert storytellers frequently use it to maintain reader engagement. Due to the intricate nature of suspense as a complex cognitive phenomenon that only emerges under the right conditions, the generation of narratives with suspense, herein referred to as *suspenseful story generation*, remains an open research challenge. Prior research pertaining to narrative suspense (O’Neill and Riedl, 2011; O’Neill and Riedl, 2014; Doust and Piwek, 2017; Wilmot and Keller, 2020; Zehe et al., 2023) predominantly focuses on the computational modeling of this cognitive phenomenon, with a very limited exploration of story suspense from the language generation perspective (Cheong and Young, 2014; Fendt and Young, 2016; Delatorre et al., 2020, 2021). General-purpose generative models such as ChatGPT (OpenAI, 2022) exhibit a poor understanding of story suspense. Model fine-tuning on suspenseful story corpora does not result in an LLM that is able to capture the latent conditions necessary for the reader to feel suspense.

Given the current widespread popularity of LLMs, how can one employ them to effectively generate suspenseful stories? A psychological theory on suspense (Gerrig and Bernardo, 1994) suggests that readers feel suspense when they believe that the protagonist is facing a negative outcome and that the quantity or quality of feasible means of escaping the impending negative outcome has become diminished. Following this theory, we propose an *iterative-prompting-based planning* technique that can effectively guide LLMs in suspenseful story generation. Specifically, we establish a character with a goal and a negative outcome if they fail. We prompt the LLM to consider the different ways the protagonist might achieve their goal. We then adversarially attack these plans by establishing conditions in the story world in which the protagonist’s plans will fail. Repeating this several times produces an outline for the story, which we then elaborate into longer text sequences (chapters). We depict and illustrate the overall intuition of our method in Figure 1.

In summary, the main contributions of our work are as follows:

- We bring together the traditional narrative & psychological theories on story suspense and the modern LLM-based language generation techniques. Our work is the first attempt at suspenseful story generation with LLMs.
- We propose an iterative-prompting-based planning technique for suspenseful story generation. This theory-grounded technique works in a fully zero-shot manner and does not rely on any supervised story corpora. We perform extensive human evaluations and in-depth analyses that demonstrate the effectiveness of our method in automatically creating suspenseful stories with the LLM.
- Using our validated method, we further advance the understanding of the factors that contribute to reader perceptions of suspense through controlled intervention studies.

2 Related Work

2.1 Automated Story Generation

Early work on story generation used symbolic planning (Meehan, 1976; Lebowitz, 1987; Cavazza et al., 2003; Porteous and Cavazza, 2009; Riedl and Young, 2010; Ware and Young, 2010; Ware and Siler, 2021). These systems require substantial knowledge engineering of logical constraints, limiting their generality, and don’t always generate plots or stories in natural language.

Neural language modeling approaches (Roemmele, 2016; Khalifa et al., 2017; Martin et al., 2018; Clark et al., 2018; Yao et al., 2019; Fan et al., 2019; Rashkin et al., 2020; Ammanabrolu et al., 2021; Jin et al., 2022) have been applied to story generation because they circumvent the need for manual knowledge engineering and tend to produce relatively fluent, varied, and naturalistic language. A significant amount of work focuses on the controllability of generators (Peng et al., 2018; Lin and Riedl, 2021; Peng et al., 2022). Others focus on improving the system from different dimensions of story goodness, e.g., commonsense reasoning (Guan et al., 2019, 2020; Ammanabrolu et al., 2021) and temporal and causal relationships (Guan et al., 2020; Ammanabrolu et al., 2021; Han et al., 2022).

Large, pre-trained language models such as GPT-3, ChatGPT, and GPT-4, are capable of generating

longer, more fluent story sequences. Generation can be extended to many thousands of words (Yang et al., 2022b,a). However, LLMs have been unreliable when it comes to generating novel, suspenseful stories. One reason for this is that suspense is a complex cognitive phenomenon that does not emerge in the latent state representations of a transformer. Instead, one must infer the cognitive state of the reader, or induce a narrative structure with known properties that co-occur with suspense.

There exist works on suspenseful story generation (Cheong and Young, 2014; Fendt and Young, 2016; Delatorre et al., 2020, 2021), but they all require substantial knowledge engineering, making them limited in generality.

2.2 Story Suspense

Suspense is a complex affective response to the events in a narrative that combines hope simultaneously with fear coupled with a cognitive state of uncertainty (Ortony et al., 1988). Why do some stories evoke feelings of suspense when others do not? Gerrig and Bernardo (1994) provide an accounting of suspense backed up by psychological studies. First, the reader must be *transported* into the fictional world. *Transportation* refers to the phenomenon of feeling immersed in the story world (Gerrig, 1993). Transportation is accompanied by a dampening of critical reasoning about facts other than those presented in the story (for example, we know that James Bond will not be killed, but we become anxious about his prospects of surviving a mission anyway). Second, the reader is given a reason to hope for an important target outcome. This could be a very positive state for the protagonist (the protagonist gets together with their love interest), or the avoidance of a very negative state (the hero is not killed). Typically, the reader develops empathy for the protagonist (Jose and Brewer, 1984). Third, the reader engages in problem-solving to figure out how the protagonist can achieve their goal (or avoid the negative state). Fourth, the space of actions the protagonist can take must have few paths to success, and/or those paths must be seen as having a low probability of success.

Another factor related to suspense is the narratological theory of *disparities of knowledge* (Brangan, 2013). This framework proposes that readers and story characters can have different knowledge of the story world. When a character knows something that the reader doesn’t know, the revelation of

that information can invoke surprise in the reader. However, when the reader knows something that the story world characters don’t know, and that knowledge is related to the protagonist’s chances of a negative outcome, then the reader may feel suspense. Thus, a key consideration in suspense generation is reasoning about what characters know versus what readers know.

O’Neill and Riedl (2014) attempt to create a computational model of the accounting of suspense (Gerrig and Bernardo, 1994). In order to determine whether a story might be perceived as suspenseful, they generate plans for the protagonist and analyze the quantity and quality of possible plans. However, since they use a symbolic planner, they must know the characters and possible actions in advance, limiting practical applicability. Their detection technique does not lend itself to story generation. Doust and Piwek (2017) propose a graph analysis approach to suspense detection. Wilmot and Keller (2020) propose a hierarchical language-modeling approach to suspense detection. Zehe et al. (2023) propose to detect suspense by training on a corpus of dangerous situations.

3 Iterative-Prompting-based Planning

In this section, we introduce our method for generating suspenseful stories with an LLM: *Iterative-Prompting-based Planning*. One of the reasons that LLMs cannot be easily prompted to generate suspenseful stories is because suspense is an affective response to a cognitive state that only comes about under certain circumstances. There must be a protagonist that is empathetic to the reader. That protagonist must face a high possibility of an outcome undesired by the protagonist and the reader. The quantity and quality (roughly in terms of expected probability) of ways of avoiding that undesired outcome are reduced. This suggests an adversarial planning process wherein the generator establishes the conditions for suspense in three stages of generation. **Background Setup** §3.1 establishes the necessary story background, describing the protagonist, their goal, and a dire situation the protagonist will be put in if the goal is not successfully achieved. **Outline Planning** §3.2 plans out the story outline by providing a series of possible actions for the protagonist to take to achieve their goal, and then generating potential reasons why actions might turn out to be ineffective in achieving the goal. In essence, we craft a plan for the pro-